Stephen Singh, MS

Location: Los Angeles, CA **Phone:** (404) 932-9019
Email: stephensingh953@gmail.com

SUMMARY

SOC FPGA Engineer with hands-on experience leading research and development at the intersection of hardware, software, and electrical engineering. Dynamic Engineer focused on leveraging the latest technologies and techniques to build and implement innovative solutions that improve programmatic control, performance, and stability for end customers.

SKILLS

- **Embedded Engineering:** Power Systems, Microcontroller Programming, PCB design, FPGA, Product Planning, Research & Development, Hardware & Circuit Design, Systems Engineering & Optimization, CAD Design & Development, Modeling & Simulation, Data Analysis, Validation, Testing, Process Control, Technical Documentation
- **Software:** Verilog, VHDL, System Verilog, Python, C/C+,+ MATLAB, VITIS, HLS, OpenCL, OpenCV, TCL, Perl, Bash, AWS, Xilinx Vivado, PetaLinux, Cadence Virtuoso, Intel Quartus, ModelSim, Git, Simulink, GEM5
- **Hardware:** Xilinx PynqZ2, Zynq-7000, Digilent Basys3, Artix-7, Xilinx MicroBlaze Arduino, ESP8266, Raspberry Pi 4
- **Leadership:** Strategic Planning & Execution, Program Development, Project Management, Process Improvement, Change Management, Cross-functional Collaboration, Stakeholder Engagement, Interdepartmental Alignment

EDUCATION

- Master of Science in Electrical & Computer Engineering, University of Florida
- Bachelor of Science in Electrical & Electronics Engineering, University of Georgia

RELEVANT EXPERIENCE

Presales System Engineer, Motorola Solutions **2018 - Present**

- Serve as the technical SME, partnering with sales staff to architect, develop, and deliver end-to-end telecommunications solutions for government, utility, transportation, first responder, and enterprise customers.
- Translated customer requirements and budget constraints into technical specifications to deploy ASTRO25, APCO P25, LTE, and 3G/4G technologies and performed RF coverage and interference analysis to meet all SLAs.
- Built RF front-ends with a focus on noise, IM distortion, harmonic suppression, and ERP according to FCC Rules.

SELECT PROJECTS

Lane-Following RC Car SoC FPGA Development

- Designed and implemented a real-time lane-following self-driving RC car on the PYNQ Z2 platform (ARM Cortex-A9 + FPGA).
- Engineered the AXI4-Stream video path, from the OV7670 into a custom RTL grayscale block and 7×7 convolution matrix IP, through AXI VDMA into PS memory while orchestrating everything live from Jupyter/MMIO.
- Developed Verilog IP cores for camera preprocessing (grayscale, 7×7 convolution) using AXI4-Stream interfacing.
- Controlled steering and throttle with servo adjustments based on lane curvature using AXI GPIO and MMIO.
- Improved accuracy by tuning the real-time lane detection algorithm (Canny edge + Hough Transform) on an ARM core and adding temporal smoothing, a confidence score, and rate-limited steering using Python/OpenCV.

AES-128 Encryption/Decryption SoC FPGA

- Engineered an AES-128 cryptographic system on Xilinx Zynq-7020 SoC to protect in-flight data and at-rest logs.
- Developed Vitis C drivers to control the AES IP over AXI-Lite and enable users to select Encrypt/Decrypt password with a 128-bit key and 128-bit block in hex via UART console and AXI-GPIO (buttons/switches/LEDs).
- Created RTL cores for AES encryption, decryption, and key expansion, using Verilog and modular FSM control for fast, confidential updates and storage with predictable latency and minimal CPU overhead.

5-Stage Pipelined RISC-V CPU

- Designed and deployed a custom 32-bit 5-stage pipelined RISC-V processor on the PYNQ-Z2 FPGA.
- Maintained CPI stability by integrating instruction and data BRAM with memory-mapped I/O for switches/LEDs.
- Accelerated integration by packaging the core as a Vivado IP and wiring it in Block Design to transform a one-off RTL into a reusable, parameterized component with standard interfaces and a managed address map.
- Achieved functional/branch coverage with no failures by verifying RTL via testbenches and waveform analysis.

HDMI Pong Game on PYNQ-Z2 with CPU-Controlled Opponent

- Led the end-to-end design, development, and implementation of a 1280×720@60Hz Pong game engine on the Zynq-7020 SoC, combining FPGA-rendered HDMI output with bare-metal C AI control via AXI-Lite.